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On L^∞ estimates for complex Monge-Ampère equations

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Citation: This paper presents a PDE-based approach to L^∞ estimates for the complex Monge-Ampère equation, covering both fixed and degenerating metrics. The method, enhancing fully non-linear equation analysis, simplifies existing pluripotential theory-based techniques.

Summary:

Auxiliary complex Monge-Ampère equations in nonlinear PDE estimates: a survey[1]

Abstract: We review recent developments in analytic estimates for solutions to fully nonlinear partial differential equations on Kähler manifolds. The comparison approach with auxiliary complex Monge-Ampère equations, introduced by the author along with Phong and Tong, has been pivotal in these developments. We also discuss the applications of these estimates in the study of the geometry of Kähler metrics.

1. Introduction

Nonlinear partial differential equations (PDEs) have attracted attention from mathematicians over the past decades, especially those with backgrounds in geometry and physics. On a Kähler manifold, the solutions to these geometric PDEs give rise to canonical metrics, such as the Calabi-Yau and Kähler-Einstein (KE) metrics. Since Yau's solution of the Calabi conjecture [**35**], the key ingredients in solving equations and understand-

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ing the asymptotic behavior of their solutions near singular sets have been the *a priori* estimates, among which the L^∞ (or C^0) estimates are of the most importance and pose the greatest challenge. To illustrate the importance of L^∞ estimates, let us consider the complex Monge-Ampère equation, which has a uniquely storied history. Early on, they were recognized as the defining difficulty in the problem of finding KE metrics. Yau's introduction [35] of Moser iteration in 1978 marked a crucial step in his solution of the Calabi conjecture, and ushered in a new era for complex MA equations. Subsequently, Kołodziej [28] obtained sharp L^∞ estimates for complex Monge-Ampère equations, which even hold for a family of Monge-Ampère equations with degenerating Kähler classes [8, 11]. Alternatively, a non-sharp L^∞ estimate was derived by Blocki [2] who applied the Aleksandrov-Bakelman-Pucci (ABP) maximum principle from PDE theory.

Kolodziej's method of proof relies on the pluripotential theory developed in the late 1970's by Bedford and Taylor [1], which is unique and however confined to complex Monge-Ampère equations only. In their work [23], the author, Phong and Tong have introduced a noval comparison method which yields sharp L^∞ estimates for a broad family of fully nonlinear PDEs on Kähler manifolds, including the complex MA and Hessian equations. This new approach is purely PDE-based, with one of the advantages being its applicability to equations with degenerating reference metrics and even non-Kähler metrics [16]. Furthermore, these methods provide a resolution to the conjectured geometric diameter bounds and uniform Sobolev inequalities of Kähler metrics, with conditions solely imposed on the integral of volume forms (c.f. [20, 21]).

2. L^∞ estimates for nonlinear partial differential equations

In this section, we review some recent progress on the L^∞ estimates to fully nonlinear partial differential equations. We begin with the classical results concerning the complex Monge-Ampère equations on a Kähler manifold.

Let (X, ω_X) be a compact Kähler manifold with complex dimension n and ω_X a given Kähler metric. The classical complex Monge-Ampère equation is defined by

$$(\omega_X + i\partial\bar\partial\varphi)^n = e^F \omega_X^n, \quad \sup_X \varphi = 0, \tag{2.1}$$

where e^F is a smooth function satisfying the compatibility condition $\int_X e^F \omega_X^n = \int_X \omega_X^n$. The equation (2.1) admits a unique smooth solution by the classical theorem of Yau [35]. The proof relies on the continuity method along with some key *a priori* estimates, among which the L^∞ estimate is the key.

Lemma 2.1. *If φ satisfies the equation (2.1), then*

$$\|\varphi\|_{L^\infty} \leq C \tag{2.2}$$

for some constant $C > 0$ depending on ω_X, n, p and $\|e^F\|_{L^p}$, for $p > 1$.

When $p > n$, Lemma 2.1 was proved by Yau [35] using the Moser iteration; when $p > 2$, it can be derived from the ABP maximum principle [2]; and the sharp case $p > 1$ was first obtained by Kolodziej [28] utilizing the pluripotential theory. We give a very sketched proof of Kolodziej. To begin with, we recall the uniform integrability of quasi-plurisubharmonic (PSH) functions on X (see [27, 32]).

Lemma 2.2. *There are constants $\alpha > 0$ and $C > 0$ depending only on n, ω_X such that*

$$\int_X e^{-\alpha\psi}\omega_X^n \le C, \quad \forall \psi \in PSH(X,\omega_X) \text{ with } \sup \psi = 0.$$

[Sketched proof of Lemma 2.1] We follow the approach in [28]. Define the capacity of a Borel set E as

$$Cap_{\omega_X}(E) = \sup\Big\{ \int_K (\omega_X + i\partial\bar\partial u)^n : \ u \in PSH(X,\omega_X),\ 0 \le u \le 1\Big\}.$$

Here $(\omega_X + i\partial\bar\partial u)^n$ is defined as a measure by the theory of Bedford-Taylor [1]. We claim that for any $s > 0$,

$$r^n Cap_{\omega_X}(\{\varphi < -s - r\}) \le C(Cap_{\omega_X}(\{\varphi < -s\}))^{1+\delta}, \quad \forall r \in [0,1], \tag{2.3}$$

for some constants C and $\delta > 0$. Once this is obtained, an iteration argument yields that $\{\varphi < -S_\infty\} = \emptyset$ for some large constant S_∞, which will finish the proof. For any given $u \in PSH(X,\omega_X)$ with $0 \le u \le 1$. By the comparison principle for bounded and PSH functions, we have

$$\begin{aligned} r^n \int_{\{\varphi<-s-r\}} (\omega_X + i\partial\bar\partial u)^n &\le \int_{\{\varphi<-s-r+ru\}} (r\omega_X + ri\partial\bar\partial u)^n \\ &\le \int_{\{\varphi<-s\}} (\omega_X + i\partial\bar\partial\varphi)^n. \end{aligned} \tag{2.4}$$

We define the extremal function V_s associated to the subset $\Omega_s := \{\varphi < -s\}$

$$V_s = \sup\{u : \ u \in PSH(X,\omega_X),\ u \le 0 \text{ in } \Omega_s\}.$$

V_s satisfies the properties $V_s \in L^\infty \cap PSH(X,\omega_X)$. We denote $M_s := \sup_X V_s$ and calculate

$$\begin{aligned} \int_{\{\varphi<-s\}} (\omega_X + i\partial\bar\partial\varphi)^n &\le e^{-\alpha M_s/p^*} \int_{\Omega_s} e^F e^{-\alpha(V_s - M_s)/p^*}\omega_X^n \\ &\le e^{-\alpha M_s/p^*}\Big(\int_X e^{pF}\omega_X\Big)^{1/p}\Big(\int_X e^{-\alpha(V_s-M_s)}\omega_X^n\Big)^{1/p^*} \\ &\le Ce^{-\alpha M_s/p^*}, \end{aligned} \tag{2.5}$$

where $p^* = \frac{p}{p-1}$ and we have used the uniform integrability of the PSH function $V_s - M_s$. Moreover, it follows from the equation $(\omega_X + i\partial\bar\partial V_s)^n = 0$

in $X\backslash\overline{\Omega_s}$, that $M_s + 1 \geq (Cap_{\omega_X}(\Omega_s))^{-1/n}$. Combining this with (2.4) and (2.5) we obtain (2.3) by a calculus inequality.

The result of Lemma 2.1 can be generalized to a family of complex Monge-Ampère equations (see [8, 11]). More precisely, let χ be a semi-positive smooth closed $(1,1)$-form. We consider the family of Kähler metrics

$$\omega_t = \chi + t\omega_X, \quad t \in (0,1], \tag{2.6}$$

and the family of complex Monge-Ampère equations with ω_t the background metrics

$$(\omega_t + i\partial\bar{\partial}\varphi_t)^n = c_t e^F \omega_X^n, \quad \sup_X \varphi_t = 0. \tag{2.7}$$

Here $c_t = \frac{\int_X \omega_t^n}{\int_X \omega_X^n} > 0$ is the normalizing constant. By an analogous argument as in the proof of Lemma 2.1, the estimate (2.2) holds uniformly (i.e. independent of t) for φ_t.

As we have seen, this approach relies heavily on the pluripotential theory and is confined to the quasi-plurisubharmonic functions. To study more general partial differential equations, it is necessary to introduce new methods on these analytic estimates. We next explain the comparison approach in [23].

2.1. L^∞ estimates with auxiliary complex Monge-Ampère equations. We consider a nonlinear operator f which satisfies the conditions:

1. $f = f(\lambda) : \Gamma \to \mathbb{R}_+$ is homogeneous of degree one and is symmetric on an admissible symmetric cone $\Gamma \subset \mathbb{R}^n$; it is necessary to assume $\Gamma_n \subset \Gamma \subset \Gamma_1$ (c.f. [5]), where Γ_k is the k-th symmetric cone in $\mathbb{R}^n$;
2. $\partial f/\partial\lambda_i > 0$ for all $\lambda \in \Gamma$, which amounts to the ellipticity of equation (2.9);

   additionally, we require the following structural condition on f:
3. for some constant $\gamma > 0$, it holds that

$$\det\left(\frac{\partial f(\lambda[h])}{\partial h_{ij}}\right) \geq \gamma, \quad \text{for all } \lambda[h] \in \Gamma, \tag{2.8}$$

   where $\lambda[h]$ denotes a vector of eigenvalues of the endomorphism $h : \mathbb{R}^n \to \mathbb{R}^n$.

With the same notations as above, we consider the following fully nonlinear partial differential equation on X

$$f(\lambda[h_\varphi]) = e^{F/n}, \quad \sup\varphi = 0, \tag{2.9}$$

where F is a smooth function normalized by $\int_X e^F\omega_X^n = \int_X \omega_X^n$, $h_\varphi : TX \to TX$ is the endmorphism defined by the $(1,1)$-form $\omega_\varphi = \omega_X + i\partial\bar{\partial}\varphi$ relative to ω_X.

We remark that geometric partial differential equations such as complex Monge-Ampère and σ_k-Hessian equations all satisfy these structural conditions. More generally, Harvey-Lawson [25, 26] recently show that any Garding-Dirichlet polynomial satisfies (2.8), thus providing a wide class of nonlinear equations for which Theorem 2.3 applies.

Theorem 2.3 ([23]). *Suppose φ is a solution to (2.9) and $q > n$ is a constant. Then we have*

$$\|\varphi\|_{L^\infty} \leq C, \tag{2.10}$$

for some constant $C >$ that depends on n, q, γ, ω_X and the q-th entropy of e^F

$$\mathrm{Ent}_q(F) = \int_X e^F |F|^q \omega_X^n.$$

The main idea in the proof of Theorem 2.3 is to introduce auxiliary complex Monge-Ampère equations with which we compare the solution φ. Given the uniform integrability of quasi-PSH equations (see Lemma 2.2), we obtain a *free* integral control of φ, an idea that goes back to Chen-Cheng [6] in their study of constant scalar Kähler metrics. We now give an outlined proof of Theorem 2.3. [Outlined proof of Theorem 2.3] We break the proof into four steps. Let φ be the solution to (2.9). As above, let $\Omega_s = \{\varphi < -s\}$ be a sub-level set of φ for $s > 0$.

Step 1. We consider the following auxiliary complex Monge-Ampère equation associated with φ

$$(\omega_X + i\partial\bar\partial\psi_{s,k})^n = \frac{\tau_k(-\varphi - s)}{A_{s,k}} e^F \omega_X^n, \quad \sup \psi_{s,k} = 0, \tag{2.11}$$

where $\tau_k(\cdot) > 0$ is a smoothing of $x \cdot \chi_{\mathbb{R}_{\geq 0}}(x)$, and as $k \to \infty$

$$A_{s,k} = \frac{1}{V}\int_X \tau_k(-\varphi - s) e^F \omega_X^n \to A_s = \frac{1}{V}\int_{\Omega_s} (-\varphi - s) e^F \omega_X^n$$

For each s and k, $\psi_{s,k}$ exists and is smooth by Yau's theorem [35].

Step 2. We compare $\varphi + s$ with the solution $\psi_{s,k}$ of the auxiliary equation.

We introduce a test function

$$\Phi := -\varepsilon(-\psi_{s,k} + \Lambda)^{\frac{n}{n+1}} - (\varphi + s)$$

for suitable constants

$$0 < \varepsilon := \Big(\frac{n+1}{n^2}\Big)^{\frac{n}{n+1}} A_{s,k}^{\frac{1}{n+1}} \gamma^{-\frac{1}{n+1}}, \quad 0 < \Lambda := \frac{1}{(n+1)n^{n-1}} \frac{A_{s,k}}{\gamma}. \tag{2.12}$$

Apply the maximum principle to the function Φ by calculating

$$\Delta_G \Phi \geq c(n)\varepsilon\gamma^{1/n}(-\psi_{s,k} + \Lambda)^{-\frac{1}{n+1}} \Big(\frac{-\varphi - s}{A_{s,k}}\Big)^{1/n} - 1,$$

It is in this inequality where we need the structural condition (2.8) on $f(\lambda[\cdot])$. Here $\Delta_G \Phi := G^{i\bar j} \frac{\partial^2 \Phi}{\partial z_i \partial \bar z_j}$ is the linearized operator of $\log f(\lambda[h])$ at $h = h_\varphi$, where $G^{i\bar j} = \frac{\partial}{\partial h_{ij}} \log f(\lambda[h])$.

Together with the choice of ε, Λ in (2.12) yields that $\Phi \leq 0$.

Step 3. $\Phi \leq 0$ implies in the open set Ω_s (which we assume to be nonempty)

$$\Big(\frac{-\varphi - s}{A_{s,k}}\Big)^{(n+1)/n} \leq C(n)(-\psi_{s,k} + A_{s,k}).$$

Exponentiating and integrating this over Ω_s yields the Trudinger-type inequality

(2.13)
$$\int_{\Omega_s} \exp\Big[\beta\Big(\frac{-\varphi - s}{A_{s,k}}\Big)^{(n+1)/n}\Big]\omega^n \leq e^{\beta C(n) A_{s,k}} \int_X e^{-\beta C(n)\psi_{s,k}} \leq C(n, E(\varphi)).$$

Note that we have used $A_{s,k} \leq E(\varphi) := \int_X (-\varphi) e^F \omega_X^n$, and the latter integral is uniformly bounded if $\beta > 0$ is chosen and fixed such that $\beta C(n) < \alpha(X, \omega_X)$, a constant in the uniform exponentially integrality of ω_X-PSH functions (see Lemma 2.2).

With (2.13) at hand, we apply the Hölder's and Young's inequalities to get

$$A_s \leq C\Big(\int_{\Omega_s} e^F \omega_X^n\Big)^{1+\delta}$$

where $\delta = \frac{q-n}{nq} > 0$ is a constant, and C depends additionally on $E(\varphi)$. This easily yields that the monotone function $\phi(s) := \int_{\Omega_s} e^F \omega_X^n$ satisfies the decay estimate

$$r\phi(s + r) \leq C(\phi(s))^{1+\delta_0}, \quad \forall s, r > 0.$$

A De Giorgi-type iteration argument as used by Kołodziej [28] implies $\phi(s) = 0$ for $s >> 1$. Hence $\Omega_s = \emptyset$ for $s >> 1$.

Step 4. It only remains to remove the dependence of the constant on $E(\varphi)$. For complex Monge-Ampère equations, this follows easily from the Jensen's inequality applied to the inequality in Lemma 2.2. For the general nonlinear partial differential equations f satisfying the above conditions, we utilize the same $\psi_{t,k}$ of the auxiliary equation further and argue as in [18] to derive the weighted L^p bound of φ by certain entropy of e^F. We refer readers to the [18] for more details.

We remark that with a slight modification of the proof, the L^∞ estimate in Theorem 2.3 holds uniformly for a family of differential equations (2.9) when ω_φ is replaced by $\omega_t + i\partial\bar\partial\varphi_t$ for $t \in (0, 1]$ and ω_t as defined in (2.6). This method of utilizing auxiliary equations turns out to be flexible and has been applied to many problems, including the stability estimates for solutions to complex Monge-Ampère and Hessian equations ([24]) and deformed Hermitian-Yang-Mills equations ([30]), and the L^∞ estimates for parabolic Monge-Ampère and Hessian equations ([7, 36]). The idea has also been extended to deal with equations on non-Kähler manifolds in [18], which we now briefly discuss in the next subsection.

2.2. L^∞ estimates on complex Hermitian manifolds. If (X, ω_X) is a non-Kähler Hermtian manifold, the obvious adaption of the argument in the proof of Theorem 2.3 fails, mainly due to the non-solvability of the auxiliary equations (2.21). Though by the theorem of Tosatti-Weinkove [33],

a complex Monge-Ampère equation

$$(\omega_X + i\partial\bar\partial\varphi)^n = e^F\omega_X^n, \quad \sup\varphi = 0$$

admits a unique solution when the function F on the right-hand side is replaced by $F + b$ for some *unknown* constant $b \in \mathbb{R}$, no bound on b poses the serious difficulty in applying the argument in the proof of Theorem 2.3. To overcome this, in [18], we introduce a *local* comparison technique. The upshot is that if φ is a smooth solution to (2.9) on the Hermitian manifold (X, ω_X), the same conclusion as that in Theorem 2.3 remains valid. We outline the main ideas below.

It suffices to prove a lower bound of φ in X. Assume that $x_0 \in X$ is a minimum point of φ, and without loss of generality, we assume that in a local coordinate chart $\Omega := B_{\mathbb{C}^n}(x_0, 2r_0)$ centered at x_0, we have $\frac{1}{2}\delta_{ij} \le (\omega_X)_{i\bar j} \le 2\delta_{ij}$. For each $s \in (0, 2r_0^2)$, we consider

$$u_s(z) = \varphi(z) - \varphi(x_0) + \frac{1}{2}|z|^2 - s.$$

It is clear that $u_s > 0$ on $\partial\Omega$. Set $\Omega_s = \{x \in \Omega : \ u_s(x) < 0\} \Subset \Omega$. We solve the following Dirichlet boundary valued Monge-Ampère equation (which admits a smooth solution by [4])

$$(i\partial\bar\partial\psi_{s,k})^n = \frac{\tau_k(-u_s)}{A_{s,k}} e^F\omega_X^n \quad \text{in } \Omega, \quad \psi_{s,k} = 0 \text{ on } \partial\Omega, \tag{2.14}$$

with $i\partial\bar\partial\psi_{s,k} > 0$ in Ω. Here $A_{s,k} = \int_\Omega \tau_k(-u_s)e^F\omega_X^n$ is chosen so that $\int_\Omega (i\partial\bar\partial\psi_{s,k})^n = 1$. It follows then from a local version of Lemma 2.2 that (c.f. [28])

$$\int_\Omega e^{-\alpha\psi_{s,k}}\omega_X^n \le C, \quad \forall s, k, \tag{2.15}$$

for some uniform constants $\alpha, C > 0$.

Consider the test function

$$\Psi = -\varepsilon(-\psi_{s,k})^{\frac{n}{n+1}} - u_s$$

with the constant $\varepsilon > 0$ given by $\varepsilon^{n+1} = A_{s,k}\frac{(n+1)^n}{\gamma n^{2n}}$. We can show that $\Psi \le 0$ in Ω by applying the maximum principle. This inequality combined with (2.15) yields an upper bound of $-\varphi(x_0)$. Details are referred to [18].

We remark that on a Hermitian manifold, the L^∞ estimates have been known for the complex Monge-Ampère equations by the Moser iteration [33], and for the Hessian equations by adapting an analogue of pluripotential theory [9, 10]. The technique of local comparison has also been applied to fully nonlinear partial differential equations on complex manifolds with boundary, and *almost* complex manifolds in [18]. It constitutes a powerful De Giorgi-Nash-Moser theory for fully non-linear equations, particularly in complex geometry.

In case the structural condition (2.8) is not met, it is well-known ([31, 13]) that one needs certain subsolutions to get effective estimates. We next

outline another application of the approach of auxiliary equations in deriving uniform L^∞ estimates assuming the existence of $\mathcal{C}$-subsolution introduced in [31].

2.3. L^∞ estimates when sub-solutions exist. In this subsection, we assume $f : \Gamma \to \mathbb{R}$ is a nonlinear operator satisfying the condition (2) in subsection 2.1, but not necessarily the structural condition (2.8).

Let χ be a given closed $(1,1)$-form on X, and ω be a Kähler metric in the set

$$\mathcal{W}(A, K, p, \omega_X) := \{\omega \text{ Kähler} : \ [\omega] \cdot [\omega_X]^{n-1} \leq A, \ \mathrm{Ent}_p(F_\omega) \leq K\}, \tag{2.16}$$

where $A, K, p > n$ are given numbers, and F_ω is the *relative volume* of ω defined by

$$e^{F_\omega} = \frac{\omega^n/[\omega]^n}{\omega_X^n/[\omega_X]^n}.$$

In the equation (2.9), we let $h_\varphi := \omega^{-1} \cdot (\chi + i\partial\bar\partial\varphi)$ be the endomorphism defined by $\chi + i\partial\bar\partial\varphi$ relative to ω. To derive uniform L^∞ estimates for solutions to (2.9) without (2.8), we first recall the following notion of subsolution in [31] (for the parabolic version, see [29]).

Definition 2.4. *Given $\delta > 0, R > 0$, we say a $(1,1)$-form $\chi' = \chi - i\partial\bar\partial\theta$ is a (δ, R)-subsolution of the equation (2.9), if the following holds*

$$(\lambda[h_{\chi'}] - \delta\mathbf{1} + \Gamma_n) \cap \partial\Gamma^{e^{F(z)/n}} \subset B(0, R) \subset \mathbb{R}^n, \quad \forall z \in X, \tag{2.17}$$

where $\mathbf{1} = (1, \ldots, 1) \in \mathbb{R}^n$ is the fixed vector, $h_{\chi'} = \omega^{-1} \cdot \chi'$, and

$$\Gamma^{e^{F(z)/n}} = \{\lambda \in \Gamma \mid f(\lambda) \geq e^{F(z)/n}\}.$$

We assume a a (δ, R)-subsolution $\chi' = \chi - i\partial\bar\partial\theta$ satisfies

$$-n\kappa_1 \leq \mathrm{tr}_\omega \chi' \leq \kappa_2, \tag{2.18}$$

for given constants $\kappa_1 \geq 0$, $\kappa_2 \geq 0$. We then have the following theorem:

Theorem 2.5 ([17])**.** *Let φ is a solution to the equation (2.9), then there exists a constant $C > 0$ depending on n, A, K, p, and $\delta, R, \kappa_1, \kappa_2$ such that*

$$\sup_X |(\varphi + \theta) - \sup_X(\varphi + \theta)| \leq C. \tag{2.19}$$

When ω is a fixed metric, this theorem can also be proved by applying the ABP maximum principle ([31]). The novelty of the approach of auxiliary equations is that it gives uniform estimates when the reference ω varies in the set $\mathcal{W}(A, K, p, \omega_X)$. We sketch the steps involving auxiliary equations in the proof of Theorem 2.5.

First, it can be proved that any ω-PSH function ψ with $\sup \psi = 0$ satisfy the uniform integrability condition:

$$\int_X (-\psi)^q \omega^n \leq C_q, \quad \forall q > 1. \tag{2.20}$$

We denote $\varphi' = \varphi + \theta$ and may assume $\sup \varphi' = 0$. For any $s \geq 0$, we denote $\Omega_s = \{z \in X : \varphi'(z) < -s\}$. Solve the following auxiliary complex Monge-Ampère equation

$$(\omega + i\partial\bar\partial\psi_{s,k})^n = \frac{\tau_k(-\varphi' - s)}{A_{s,k}}\omega^n, \quad \sup_X \psi_{s,k} = 0, \tag{2.21}$$

where the constant $A_{s,k}$ is defined so that the equation is compatible. In this case, consider the test function

$$\Psi = -\varepsilon(-\psi_{s,k} + \Lambda)^{\frac{n}{n+1}} - \varphi' - s, \tag{2.22}$$

where

$$\varepsilon = \beta_1 A_{s,k}^{\frac{1}{n+1}}, \quad \Lambda = \beta_2 A_{s,k}, \tag{2.23}$$

where β_1, β_2 are constants depending on n, R, δ, κ_1 only. As usual, the key is to show $\Psi \leq 0$ on X, for which we need the subsolution condition. In fact, at a maximum point x_0 of Ψ, we have

$$\chi' + i\partial\bar\partial\varphi' = \chi' - \delta\omega + \rho, \tag{2.24}$$

for some *positive* $(1,1)$-form ρ. This implies that

$$\lambda[h_{\chi'+i\partial\bar\partial\varphi'}] \in \lambda[h_{\chi'}] - \delta\mathbf{1} + \Gamma_n.$$

Hence the eigenvalues of $h_{\chi'+i\partial\bar\partial\varphi'}$ are bounded by R by the definition of (R,δ)-subsolution χ'. Up to this point, an appropriate choice of the constants ε, Λ in (2.23) yields $\Psi \leq 0$. Together with the uniform estimate of $\psi_{s,k}$ as in (2.20), we can apply a similar argument as before to derive the uniform L^∞ estimate of φ'.

We remark that Theorem 2.5 remains valid on a Hermitian manifold, in which case the estimate is not uniform.

3. Applications in Kähler geometry

In this section, we briefly discuss some applications of the approach of auxiliary equations to the geometry of Kähler metrics, developed recently by the author with Phong, Song and Sturm in [**19**, **20**, **21**, **22**]. Notably, a uniform Sobolev inequality is proved for Kähler metrics satisfying only integral bounds on the volume forms, which furthermore allows us to build a general theory of global geometric analysis on singular Kähler spaces. It has played a role in the study of Hermitian-Einstein connections on singular space [**3**] and the Bogomolov-Gieseker inequality on singular Kähler threefolds [**15**].

As in the preceding section, let (X, ω_X) be a closed Kähler manifold of dimension n. Given the parameters $A, K, p > n$, we define a subset $\mathcal{W} := \mathcal{W}(A, K, p, \omega_X)$ of Kähler metrics on X by (2.16). We are interested in the geometry of $\omega \in \mathcal{W}$, like its the diameter and noncollapsing properties. These are known when assuming the Ricci curvature lower bound of ω ([**12**]). However, many problems in Kähler geometry do not meet such an assumption, for example, the Kähler metrics along a Kähler-Ricci flow. With

the aid of uniform estimates on the Green's function, we can understand the geometry of ω in a better manner.

We recall the Green's function G_ω of $\omega \in \mathcal{W}$:

$$\Delta_{\omega,y} G_\omega(x,y) = -\delta_x(y) + \frac{1}{V_\omega}, \quad \int_X G_\omega(x,y)\omega_y^n = 0,$$

where $V_\omega = \int_X \omega^n$. G_ω is symmetric and smooth away from the diagonal of $X \times X$. We have the following effective estimates on the Kähler metric $\omega \in \mathcal{W}$.

Theorem 3.1 ([19, 20, 21, 22]). *Given $p > n$, $A > 0$, $K > 0$, there is a constant $C = C(n, \omega_X, p, A, K) > 0$, $\epsilon = \epsilon(n,p) > 0$ s.t. for any $\omega \in \mathcal{W}(A, K, p, \omega_X)$, the following hold:*

1. *integral estimates on the Green's function:*
$$\sup_{x\in X} \int (|G_\omega(x,\cdot)| + V_\omega^\epsilon |\nabla G_\omega(x,\cdot)|_\omega^{1+\epsilon})\omega^n + \sup_{y\in X}(-G_\omega(x,y))V_\omega \le C$$
2. *diameter bound:* $\operatorname{diam}(X,\omega) \le C$
3. *non-collapsing: there is an $\alpha = \alpha(n,p) > 0$ such that for any $R \in (0,1)$*
$$\operatorname{Vol}_\omega(B_\omega(x,R)) \cdot {V_\omega}^{-1} \ge C^{-1} R^\alpha,$$
where $B_\omega(x,R)$ denotes the geodesic ball with respect to the metric ω, centered at x of radius R.
4. *Sobolev inequality: for some $q = q(n,p) > 1$, such that $\forall u \in W^{1,2}$*
$$\Big(\frac{1}{V_\omega}\int_X |u-\bar u|^{2q}\omega^n\Big)^{\frac{1}{q}} \le \frac{C}{V_\omega}\int_X |\nabla u|_\omega^2 \omega^n,$$
where $\bar u = \frac{1}{V_\omega}\int_X u\omega^n$ is the average of u on (X,ω).

The diameter and noncollapsing estimates have been obtained in [34] using a different method. In the original works, an additional assumption on non-degeneracy of e^{F_ω} was made, but this has been removed by [22] and [14] independently. Theorem 3.1 can be applied to short-time and long-time Kähler-Ricci flows, providing the crucial (pre)compactness of Kähler metrics along these flows. It is also extended to certain singular normal Kähler varieties in [21], and on these space, we also have uniform heat kernel estimates and spectral estimates.

A key ingredient in the proof of Theorem 3.1 is the following mean-value inequality.

Lemma 3.2 (Mean-value inequality). *Given $\omega \in \mathcal{W}(A,K,p,\omega_X)$, suppose $v \in L^1(X,\omega^n)$ satisfies $\int_X v\omega^n = 0$ and*

$$v \in C^2(\overline{\Omega_0}), \quad \Delta_\omega v \ge -1 \text{ in } \Omega_0$$

where $\Omega_s = \{v > s\}$. Then there is a constant $C = C(\omega_X, A, p, K) > 0$ such that

$$\sup_X v \le C(1 + \frac{1}{V_\omega}\int_X |v|\omega^n).$$

The proof of this Lemma follows a similar line as in the proof of Theorem 2.3. The auxiliary complex Monge-Ampère equations one can use are

$$(\theta + i\partial\bar\partial\psi_{s,k})^n = V_\omega \frac{\tau_k(v-s)}{A_{s,k}} e^{F_\omega}\omega_X^n, \quad \sup_X \psi_{s,k} = 0,$$

where $\theta \in [\omega]$ is a smooth closed $(1,1)$-form which is bounded in $C^2(X,\omega_X)$, and $A_{s,k}$ is a constant chosen so that (3) is compatible. Consider a test function

$$\Phi := -\varepsilon(-\psi_{s,k} + \varphi + \Lambda)^{\frac{n}{n+1}} + (v-s),$$

where φ a Kähler potential of ω relative to θ with $\sup_X \varphi = 0$. One can prove that $\Phi \le 0$ on X, from which the upper bound of v follows similarly.

Applying Lemma 3.2 to the function $v := -V_\omega \cdot G_\omega(x,\cdot)$ gives an upper bound of $-G_\omega$ in terms of $\|G_\omega(x,\cdot)\|_{L^1(X,\omega)}$. The latter being bounded is proved in [**22**].

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